resids_loo

Taken from Peter S

Load Data

First Analysis

What are the different county variables that effect the conviction rates of different defendant types

Population

Log.pop Log.urban #Log.rural <- correlated with urban Young.Males.Population Largest.Municipality.Population High.School.Graduates

Racial/Ethnicity

White.Population <- baseline African.American.Population Native.American.Alaskan.Population Asian.or.Pacific.Islander.Population # Other.Race.Population <- sum to 1 so remove basline Hispanic.or.Latino.Population

Econ

#Single.Female.Headed.Households.with.Children Unemployment.Rate #Non.Citizens
Below.Poverty.Line Median.Household.Income
#Average.Commute.Minutes #Residential.Mobility

Crim Justice

Number.of.Criminal.Court.Judges Police.Officers.per.100.000.Residents # Law.Enforcement.Agencies.Reporting.to.UCR # Total.Number.of.Law.Enforcement.Agencies <- tied to county size, population Number.of.Full.Time.Prosecutors #Number.of.Full.Time.Sworn.Law.Enforcement.Officers < correlated with # full-time prosecutors, scales with pop # Number.of.Part.Time.Prosecutors

```
##                             Log.pop              Young.Males.Population
##                            6.548696                            4.412262
##     Largest.Municipality.Population          African.American.Population
##                            3.271584                            2.323363
##    Native.American.Alaskan.Population Asian.or.Pacific.Islander.Population
##                            1.653503                            4.343896
##        Hispanic.or.Latino.Population                  Below.Poverty.Line
##                            2.150730                            3.277401
##                   Unemployment.Rate      Number.of.Criminal.Court.Judges
##                            2.430709                            3.780157
## Police.Officers.per.100.000.Residents     Number.of.Full.Time.Prosecutors
```

```
##                                 6.611345                                4.746244
##                                  Log.pop                 Young.Males.Population
##                                 6.548696                                4.412262
##        Largest.Municipality.Population              African.American.Population
##                                 3.271584                                2.323363
##     Native.American.Alaskan.Population  Asian.or.Pacific.Islander.Population
##                                 1.653503                                4.343896
##          Hispanic.or.Latino.Population                       Below.Poverty.Line
##                                 2.150730                                3.277401
##                      Unemployment.Rate          Number.of.Criminal.Court.Judges
##                                 2.430709                                3.780157
## Police.Officers.per.100.000.Residents          Number.of.Full.Time.Prosecutors
##                                 6.611345                                4.746244
##                                  Log.pop                 Young.Males.Population
##                                 6.548696                                4.412262
##        Largest.Municipality.Population              African.American.Population
##                                 3.271584                                2.323363
##     Native.American.Alaskan.Population  Asian.or.Pacific.Islander.Population
##                                 1.653503                                4.343896
##          Hispanic.or.Latino.Population                       Below.Poverty.Line
##                                 2.150730                                3.277401
##                      Unemployment.Rate          Number.of.Criminal.Court.Judges
##                                 2.430709                                3.780157
## Police.Officers.per.100.000.Residents          Number.of.Full.Time.Prosecutors
##                                 6.611345                                4.746244
##                                  Log.pop                 Young.Males.Population
##                                 9.796998                                1.489015
##        Largest.Municipality.Population              African.American.Population
##                                 4.456119                                4.118634
##     Native.American.Alaskan.Population  Asian.or.Pacific.Islander.Population
##                                 1.531395                                5.143580
##          Hispanic.or.Latino.Population                       Below.Poverty.Line
##                                 2.431622                                3.735301
##                      Unemployment.Rate          Number.of.Criminal.Court.Judges
##                                 2.399152                                4.089678
## Police.Officers.per.100.000.Residents          Number.of.Full.Time.Prosecutors
##                                 7.630821                                7.451482
##                                Log.pop                Young.Males.Population
##                               2.263524                              2.194946
##            African.American.Population    Native.American.Alaskan.Population
##                               2.340194                              1.242947
## Asian.or.Pacific.Islander.Population         Hispanic.or.Latino.Population
##                               3.688240                              1.096280
##                     Below.Poverty.Line                     Unemployment.Rate
##                               1.924567                              1.891976
##                                Log.pop                Young.Males.Population
##                               2.304757                              2.261650
##            African.American.Population    Native.American.Alaskan.Population
##                               2.375281                              1.225758
## Asian.or.Pacific.Islander.Population         Hispanic.or.Latino.Population
##                               3.607432                              1.130524
```

```
##                  Below.Poverty.Line                  Unemployment.Rate
##                            1.944445                           1.905171
##                             Log.pop               Young.Males.Population
##                            2.304757                           2.261650
##          African.American.Population   Native.American.Alaskan.Population
##                            2.375281                           1.225758
## Asian.or.Pacific.Islander.Population         Hispanic.or.Latino.Population
##                            3.607432                           1.130524
##                  Below.Poverty.Line                  Unemployment.Rate
##                            1.944445                           1.905171
##                             Log.pop               Young.Males.Population
##                            2.854214                           3.333204
##          African.American.Population   Native.American.Alaskan.Population
##                            3.102031                           1.145490
## Asian.or.Pacific.Islander.Population         Hispanic.or.Latino.Population
##                            3.509902                           1.083678
##                  Below.Poverty.Line                  Unemployment.Rate
##                            4.043724                           2.699432
##                              Log.pop                Young.Males.Population
##                             7.508138                              2.535723
##      Largest.Municipality.Population           African.American.Population
##                             3.314580                              1.392823
##   Native.American.Alaskan.Population  Asian.or.Pacific.Islander.Population
##                             1.917264                              4.265724
##        Hispanic.or.Latino.Population                    Below.Poverty.Line
##                             2.064378                              2.660834
##                    Unemployment.Rate Police.Officers.per.100.000.Residents
##                             1.535178                              1.879753
##                              Log.pop                Young.Males.Population
##                             6.912436                              2.611277
##      Largest.Municipality.Population           African.American.Population
##                             2.921528                              1.390874
##   Native.American.Alaskan.Population  Asian.or.Pacific.Islander.Population
##                             1.909346                              5.067391
##        Hispanic.or.Latino.Population                    Below.Poverty.Line
##                             1.587196                              2.722849
##                    Unemployment.Rate Police.Officers.per.100.000.Residents
##                             1.555054                              1.878611
##                              Log.pop                Young.Males.Population
##                             7.418723                              2.795172
##      Largest.Municipality.Population           African.American.Population
##                             3.008236                              1.349511
##   Native.American.Alaskan.Population  Asian.or.Pacific.Islander.Population
##                             1.922732                              5.250425
##        Hispanic.or.Latino.Population                    Below.Poverty.Line
##                             1.681197                              2.852191
##                    Unemployment.Rate Police.Officers.per.100.000.Residents
##                             1.709256                              1.928338
##                              Log.pop                Young.Males.Population
##                             7.779984                              2.645515
##      Largest.Municipality.Population           African.American.Population
```

```
##                             2.737722                               1.330914
##    Native.American.Alaskan.Population  Asian.or.Pacific.Islander.Population
##                             1.906912                               5.368473
##         Hispanic.or.Latino.Population                      Below.Poverty.Line
##                             1.678717                               2.741535
##                     Unemployment.Rate Police.Officers.per.100.000.Residents
##                             1.625388                               1.915784
```

Summary Tables

```r
summary(lm_pd_PA)
```

```
##
## Call:
## lm(formula = car::logit(Public.Defender.conviction.rate) ~ Log.pop +
##     Young.Males.Population + Largest.Municipality.Population +
##     African.American.Population + Native.American.Alaskan.Population +
##     Asian.or.Pacific.Islander.Population + Hispanic.or.Latino.Population +
##     Below.Poverty.Line + Unemployment.Rate + Number.of.Criminal.Court.Judges +
##     Police.Officers.per.100.000.Residents + Number.of.Full.Time.Prosecutors,
##     data = rdf_PA, na.action = na.omit)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.08594 -0.24899  0.01795  0.20634  0.99201
##
## Coefficients:
##                                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)                            1.106e+00  1.387e+00   0.797    0.430
## Log.pop                                7.076e-02  1.468e-01   0.482    0.632
## Young.Males.Population                -7.150e-02  7.418e-02  -0.964    0.341
## Largest.Municipality.Population       -1.804e-07  2.763e-06  -0.065    0.948
## African.American.Population           -6.009e-03  2.335e-02  -0.257    0.798
## Native.American.Alaskan.Population     1.184e+00  1.187e+00   0.998    0.324
## Asian.or.Pacific.Islander.Population   3.417e-02  1.001e-01   0.341    0.735
## Hispanic.or.Latino.Population         -2.010e-02  3.076e-02  -0.653    0.517
## Below.Poverty.Line                     4.948e-02  4.077e-02   1.214    0.232
## Unemployment.Rate                     -1.025e-01  7.696e-02  -1.332    0.190
## Number.of.Criminal.Court.Judges        9.328e-03  6.482e-03   1.439    0.158
## Police.Officers.per.100.000.Residents -4.266e-03  4.017e-03  -1.062    0.295
## Number.of.Full.Time.Prosecutors        1.359e-02  1.338e-02   1.016    0.316
##
## Residual standard error: 0.4885 on 40 degrees of freedom
##   (14 observations deleted due to missingness)
## Multiple R-squared:  0.2281, Adjusted R-squared:  -0.00347
## F-statistic: 0.985 on 12 and 40 DF,  p-value: 0.4792
```

```r
summary(lm_p_PA)
```

```
##
## Call:
## lm(formula = car::logit(Private.conviction.rate) ~ Log.pop +
##     Young.Males.Population + Largest.Municipality.Population +
##     African.American.Population + Native.American.Alaskan.Population +
```

```
##     Asian.or.Pacific.Islander.Population + Hispanic.or.Latino.Population +
##     Below.Poverty.Line + Unemployment.Rate + Number.of.Criminal.Court.Judges +
##     Police.Officers.per.100.000.Residents + Number.of.Full.Time.Prosecutors,
##     data = rdf_PA, na.action = na.omit)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.57631 -0.27678 -0.02274  0.13805  1.27008
##
## Coefficients:
##                                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)                           -8.092e-01  1.231e+00  -0.658   0.5146
## Log.pop                                6.189e-02  1.303e-01   0.475   0.6373
## Young.Males.Population                -8.331e-02  6.582e-02  -1.266   0.2129
## Largest.Municipality.Population       -7.707e-07  2.452e-06  -0.314   0.7549
## African.American.Population           -2.853e-02  2.072e-02  -1.377   0.1762
## Native.American.Alaskan.Population     1.519e+00  1.053e+00   1.443   0.1569
## Asian.or.Pacific.Islander.Population   5.944e-02  8.879e-02   0.669   0.5071
## Hispanic.or.Latino.Population         -1.461e-02  2.730e-02  -0.535   0.5955
## Below.Poverty.Line                     7.641e-02  3.618e-02   2.112   0.0410 *
## Unemployment.Rate                      2.477e-02  6.828e-02   0.363   0.7188
## Number.of.Criminal.Court.Judges        1.379e-02  5.752e-03   2.397   0.0213 *
## Police.Officers.per.100.000.Residents -2.119e-03  3.565e-03  -0.594   0.5556
## Number.of.Full.Time.Prosecutors        6.737e-03  1.187e-02   0.567   0.5736
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4335 on 40 degrees of freedom
##   (14 observations deleted due to missingness)
## Multiple R-squared:  0.3166, Adjusted R-squared:  0.1116
## F-statistic: 1.544 on 12 and 40 DF,  p-value: 0.1488
```

```r
summary(lm_ca_PA)
```

```
##
## Call:
## lm(formula = car::logit(Court.Appointed.conviction.rate) ~ Log.pop +
##     Young.Males.Population + Largest.Municipality.Population +
##     African.American.Population + Native.American.Alaskan.Population +
##     Asian.or.Pacific.Islander.Population + Hispanic.or.Latino.Population +
##     Below.Poverty.Line + Unemployment.Rate + Number.of.Criminal.Court.Judges +
##     Police.Officers.per.100.000.Residents + Number.of.Full.Time.Prosecutors,
##     data = rdf_PA, na.action = na.omit)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.94113 -0.19062 -0.07407  0.25017  1.81307
##
## Coefficients:
##                                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)                            4.766e+00  1.925e+00   2.476   0.0176 *
## Log.pop                               -1.994e-01  2.037e-01  -0.979   0.3337
## Young.Males.Population                -2.178e-02  1.030e-01  -0.212   0.8335
## Largest.Municipality.Population        2.216e-06  3.836e-06   0.578   0.5668
## African.American.Population           -1.062e-02  3.240e-02  -0.328   0.7449
```

```
## Native.American.Alaskan.Population    -1.058e+00  1.647e+00  -0.642   0.5245
## Asian.or.Pacific.Islander.Population   1.028e-01  1.389e-01   0.740   0.4637
## Hispanic.or.Latino.Population          6.471e-04  4.270e-02   0.015   0.9880
## Below.Poverty.Line                    -8.967e-03  5.659e-02  -0.158   0.8749
## Unemployment.Rate                     -5.421e-02  1.068e-01  -0.508   0.6146
## Number.of.Criminal.Court.Judges        6.524e-03  8.997e-03   0.725   0.4726
## Police.Officers.per.100.000.Residents -5.451e-03  5.576e-03  -0.978   0.3342
## Number.of.Full.Time.Prosecutors        1.655e-02  1.857e-02   0.891   0.3781
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6781 on 40 degrees of freedom
##   (14 observations deleted due to missingness)
## Multiple R-squared:  0.1597, Adjusted R-squared:  -0.09245
## F-statistic: 0.6333 on 12 and 40 DF,  p-value: 0.8012
```

```r
summary(lm_sr_PA)
```

```
##
## Call:
## lm(formula = car::logit(Self.Represented.conviction.rate) ~ Log.pop +
##     Young.Males.Population + Largest.Municipality.Population +
##     African.American.Population + Native.American.Alaskan.Population +
##     Asian.or.Pacific.Islander.Population + Hispanic.or.Latino.Population +
##     Below.Poverty.Line + Unemployment.Rate + Number.of.Criminal.Court.Judges +
##     Police.Officers.per.100.000.Residents + Number.of.Full.Time.Prosecutors,
##     data = rdf_PA, na.action = na.omit)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.9405 -0.8514 -0.0223  1.2105  2.5600
##
## Coefficients:
##                                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)                           4.642e+00  1.148e+01   0.404   0.6898
## Log.pop                              -5.112e-01  1.055e+00  -0.484   0.6329
## Young.Males.Population                2.321e-01  4.583e-01   0.506   0.6176
## Largest.Municipality.Population      -3.197e-06  1.276e-05  -0.251   0.8044
## African.American.Population          -1.217e-01  1.479e-01  -0.823   0.4194
## Native.American.Alaskan.Population   -3.954e+00  6.678e+00  -0.592   0.5598
## Asian.or.Pacific.Islander.Population -6.440e-02  4.799e-01  -0.134   0.8945
## Hispanic.or.Latino.Population        -3.681e-01  1.488e-01  -2.473   0.0216 *
## Below.Poverty.Line                   -7.439e-03  2.029e-01  -0.037   0.9711
## Unemployment.Rate                     2.576e-01  4.418e-01   0.583   0.5659
## Number.of.Criminal.Court.Judges       4.255e-02  2.794e-02   1.523   0.1420
## Police.Officers.per.100.000.Residents  2.500e-03  2.116e-02   0.118   0.9070
## Number.of.Full.Time.Prosecutors       8.315e-02  7.152e-02   1.163   0.2575
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.845 on 22 degrees of freedom
##   (32 observations deleted due to missingness)
## Multiple R-squared:  0.3058, Adjusted R-squared:  -0.07293
## F-statistic: 0.8074 on 12 and 22 DF,  p-value: 0.6407
```

```r
summary(lm_pd_OR)
```

```
##
## Call:
## lm(formula = car::logit(Public.Defender.conviction.rate) ~ Log.pop +
##     Young.Males.Population + African.American.Population + Native.American.Alaskan.Population +
##     Asian.or.Pacific.Islander.Population + Hispanic.or.Latino.Population +
##     Below.Poverty.Line + Unemployment.Rate, data = rdf_OR, na.action = na.omit)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.0622 -0.5115  0.2469  0.8586  2.0058
##
## Coefficients:
##                                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)                           2.064666   2.874549   0.718   0.4793
## Log.pop                               0.057959   0.274778   0.211   0.8347
## Young.Males.Population                0.108649   0.246782   0.440   0.6635
## African.American.Population           0.220826   0.453664   0.487   0.6307
## Native.American.Alaskan.Population    0.132933   0.113285   1.173   0.2517
## Asian.or.Pacific.Islander.Population -0.121327   0.269382  -0.450   0.6563
## Hispanic.or.Latino.Population        -0.070519   0.033319  -2.116   0.0444 *
## Below.Poverty.Line                   -0.048044   0.106899  -0.449   0.6570
## Unemployment.Rate                     0.009681   0.148252   0.065   0.9485
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.539 on 25 degrees of freedom
##   (2 observations deleted due to missingness)
## Multiple R-squared:  0.1933, Adjusted R-squared:  -0.06486
## F-statistic: 0.7487 on 8 and 25 DF,  p-value: 0.6491
```

```r
summary(lm_p_OR)
```

```
##
## Call:
## lm(formula = car::logit(Private.conviction.rate) ~ Log.pop +
##     Young.Males.Population + African.American.Population + Native.American.Alaskan.Population +
##     Asian.or.Pacific.Islander.Population + Hispanic.or.Latino.Population +
##     Below.Poverty.Line + Unemployment.Rate, data = rdf_OR, na.action = na.omit)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.60589 -0.18627  0.01024  0.22589  0.43976
##
## Coefficients:
##                                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)                          -1.245561   0.528882  -2.355   0.0260 *
## Log.pop                               0.124555   0.051125   2.436   0.0217 *
## Young.Males.Population               -0.005870   0.049556  -0.118   0.9066
## African.American.Population          -0.123770   0.092000  -1.345   0.1897
## Native.American.Alaskan.Population    0.030590   0.022764   1.344   0.1902
## Asian.or.Pacific.Islander.Population -0.002985   0.053438  -0.056   0.9559
## Hispanic.or.Latino.Population         0.010462   0.006747   1.551   0.1326
```

```
## Below.Poverty.Line                      0.014512   0.021517   0.674   0.5058
## Unemployment.Rate                      -0.011056   0.029883  -0.370   0.7143
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3123 on 27 degrees of freedom
## Multiple R-squared:  0.371,  Adjusted R-squared:  0.1846
## F-statistic:  1.99 on 8 and 27 DF,  p-value: 0.08672
```

```r
summary(lm_ca_OR)
```

```
##
## Call:
## lm(formula = car::logit(Court.Appointed.conviction.rate) ~ Log.pop +
##     Young.Males.Population + African.American.Population + Native.American.Alaskan.Population +
##     Asian.or.Pacific.Islander.Population + Hispanic.or.Latino.Population +
##     Below.Poverty.Line + Unemployment.Rate, data = rdf_OR, na.action = na.omit)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.5117 -0.1603 -0.0317  0.1798  0.5273
##
## Coefficients:
##                                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)                           -0.436014   0.516113  -0.845  0.40564
## Log.pop                                0.147776   0.049891   2.962  0.00631 **
## Young.Males.Population                -0.036509   0.048360  -0.755  0.45682
## African.American.Population           -0.205416   0.089779  -2.288  0.03019 *
## Native.American.Alaskan.Population     0.023641   0.022214   1.064  0.29665
## Asian.or.Pacific.Islander.Population   0.051725   0.052148   0.992  0.33006
## Hispanic.or.Latino.Population          0.011029   0.006584   1.675  0.10545
## Below.Poverty.Line                     0.016274   0.020997   0.775  0.44504
## Unemployment.Rate                     -0.017919   0.029161  -0.614  0.54404
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3048 on 27 degrees of freedom
## Multiple R-squared:  0.4623, Adjusted R-squared:  0.303
## F-statistic: 2.902 on 8 and 27 DF,  p-value: 0.018
```

```r
summary(lm_sr_OR)
```

```
##
## Call:
## lm(formula = car::logit(Self.Represented.conviction.rate) ~ Log.pop +
##     Young.Males.Population + African.American.Population + Native.American.Alaskan.Population +
##     Asian.or.Pacific.Islander.Population + Hispanic.or.Latino.Population +
##     Below.Poverty.Line + Unemployment.Rate, data = rdf_OR, na.action = na.omit)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.2731 -0.6550  0.1130  0.8753  2.4455
##
## Coefficients:
##                                       Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)                          6.23300    6.06908   1.027  0.32467
## Log.pop                             -0.29708    0.53322  -0.557  0.58767
## Young.Males.Population               0.47517    0.38989   1.219  0.24636
## African.American.Population          0.31836    0.62908   0.506  0.62198
## Native.American.Alaskan.Population   0.21765    0.40187   0.542  0.59802
## Asian.or.Pacific.Islander.Population -0.19002   0.33106  -0.574  0.57658
## Hispanic.or.Latino.Population        0.14987    0.04755   3.152  0.00834 **
## Below.Poverty.Line                  -0.26484    0.22423  -1.181  0.26045
## Unemployment.Rate                   -0.31177    0.31332  -0.995  0.33935
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.787 on 12 degrees of freedom
##   (15 observations deleted due to missingness)
## Multiple R-squared:  0.6139, Adjusted R-squared:  0.3564
## F-statistic: 2.385 on 8 and 12 DF,  p-value: 0.08469
```

```r
summary(lm_pd_FL)
```

```
##
## Call:
## lm(formula = car::logit(Public.Defender.conviction.rate) ~ Log.pop +
##     Young.Males.Population + Largest.Municipality.Population +
##     African.American.Population + Native.American.Alaskan.Population +
##     Asian.or.Pacific.Islander.Population + Hispanic.or.Latino.Population +
##     Below.Poverty.Line + Unemployment.Rate + Police.Officers.per.100.000.Residents,
##     data = rdf_FL, na.action = na.omit)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.27098 -0.31064  0.03515  0.29330  1.91181
##
## Coefficients:
##                                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)                           3.389e+00  1.791e+00   1.892   0.0645 .
## Log.pop                              -7.368e-02  1.402e-01  -0.526   0.6016
## Young.Males.Population                5.505e-03  6.073e-02   0.091   0.9282
## Largest.Municipality.Population      -1.267e-08  1.638e-06  -0.008   0.9939
## African.American.Population          -6.127e-03  9.522e-03  -0.643   0.5230
## Native.American.Alaskan.Population    1.198e-03  1.814e-01   0.007   0.9948
## Asian.or.Pacific.Islander.Population -2.858e-02  1.416e-01  -0.202   0.8409
## Hispanic.or.Latino.Population        -1.318e-02  9.594e-03  -1.373   0.1760
## Below.Poverty.Line                   -1.819e-02  2.674e-02  -0.680   0.4995
## Unemployment.Rate                    -2.903e-02  3.746e-02  -0.775   0.4422
## Police.Officers.per.100.000.Residents -2.350e-04  1.149e-03  -0.204   0.8389
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5978 on 48 degrees of freedom
##   (8 observations deleted due to missingness)
## Multiple R-squared:  0.1922, Adjusted R-squared:  0.0239
## F-statistic: 1.142 on 10 and 48 DF,  p-value: 0.3523
```

```r
summary(lm_p_FL)
```

```
##
## Call:
## lm(formula = car::logit(Private.conviction.rate) ~ Log.pop +
##     Young.Males.Population + Largest.Municipality.Population +
##     African.American.Population + Native.American.Alaskan.Population +
##     Asian.or.Pacific.Islander.Population + Hispanic.or.Latino.Population +
##     Below.Poverty.Line + Unemployment.Rate + Police.Officers.per.100.000.Residents,
##     data = rdf_FL, na.action = na.omit)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.27922 -0.22774  0.04546  0.37419  1.56984
##
## Coefficients:
##                                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)                           2.010e+00  1.791e+00   1.123   0.2673
## Log.pop                              -6.889e-02  1.402e-01  -0.491   0.6254
## Young.Males.Population                3.821e-02  6.161e-02   0.620   0.5382
## Largest.Municipality.Population       2.791e-06  1.836e-06   1.520   0.1352
## African.American.Population          -7.514e-03  9.520e-03  -0.789   0.4339
## Native.American.Alaskan.Population   -5.620e-02  1.815e-01  -0.310   0.7582
## Asian.or.Pacific.Islander.Population -2.666e-01  1.543e-01  -1.728   0.0906 .
## Hispanic.or.Latino.Population        -5.578e-03  1.045e-02  -0.534   0.5960
## Below.Poverty.Line                   -2.221e-02  2.707e-02  -0.820   0.4161
## Unemployment.Rate                     3.397e-02  3.773e-02   0.900   0.3726
## Police.Officers.per.100.000.Residents -4.098e-04  1.150e-03  -0.357   0.7230
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5977 on 47 degrees of freedom
##   (9 observations deleted due to missingness)
## Multiple R-squared:  0.2203, Adjusted R-squared:  0.05445
## F-statistic: 1.328 on 10 and 47 DF,  p-value: 0.2437
```

```r
summary(lm_ca_FL)
```

```
##
## Call:
## lm(formula = car::logit(Court.Appointed.conviction.rate) ~ Log.pop +
##     Young.Males.Population + Largest.Municipality.Population +
##     African.American.Population + Native.American.Alaskan.Population +
##     Asian.or.Pacific.Islander.Population + Hispanic.or.Latino.Population +
##     Below.Poverty.Line + Unemployment.Rate + Police.Officers.per.100.000.Residents,
##     data = rdf_FL, na.action = na.omit)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.1543 -0.3338 -0.0236  0.3889  1.6693
##
## Coefficients:
##                                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)                           4.225e+00  2.040e+00   2.072   0.0445 *
```

```
## Log.pop                              -1.217e-01  1.607e-01  -0.758   0.4529
## Young.Males.Population               -3.404e-02  7.324e-02  -0.465   0.6445
## Largest.Municipality.Population       1.106e-06  2.525e-06   0.438   0.6637
## African.American.Population          -1.185e-02  1.037e-02  -1.143   0.2593
## Native.American.Alaskan.Population   -1.110e-01  1.934e-01  -0.574   0.5689
## Asian.or.Pacific.Islander.Population -4.977e-02  1.754e-01  -0.284   0.7780
## Hispanic.or.Latino.Population        -1.233e-02  1.196e-02  -1.031   0.3083
## Below.Poverty.Line                   -2.208e-02  3.014e-02  -0.733   0.4678
## Unemployment.Rate                    -2.988e-02  4.326e-02  -0.691   0.4936
## Police.Officers.per.100.000.Residents  3.641e-05  1.239e-03   0.029   0.9767
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6316 on 42 degrees of freedom
##   (14 observations deleted due to missingness)
## Multiple R-squared:  0.214,  Adjusted R-squared:  0.02692
## F-statistic: 1.144 on 10 and 42 DF,  p-value: 0.3545
```

```
summary(lm_sr_FL)
```

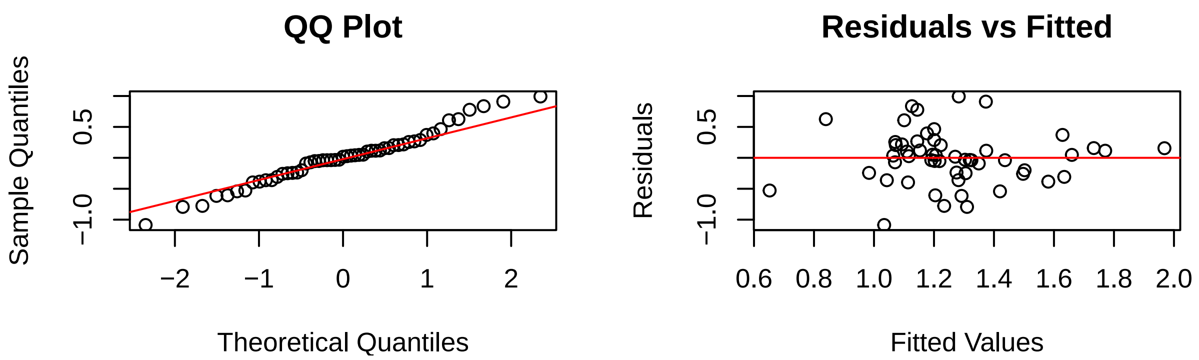
```
##
## Call:
## lm(formula = car::logit(Self.Represented.conviction.rate) ~ Log.pop +
##     Young.Males.Population + Largest.Municipality.Population +
##     African.American.Population + Native.American.Alaskan.Population +
##     Asian.or.Pacific.Islander.Population + Hispanic.or.Latino.Population +
##     Below.Poverty.Line + Unemployment.Rate + Police.Officers.per.100.000.Residents,
##     data = rdf_FL, na.action = na.omit)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.2812 -0.4362 -0.1175  0.3901  2.4856
##
## Coefficients:
##                                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)                            1.483e+00  2.509e+00   0.591   0.5575
## Log.pop                                5.188e-02  1.972e-01   0.263   0.7936
## Young.Males.Population                -1.622e-01  8.634e-02  -1.879   0.0668 .
## Largest.Municipality.Population       -2.328e-06  2.402e-06  -0.969   0.3376
## African.American.Population            1.858e-02  1.245e-02   1.492   0.1428
## Native.American.Alaskan.Population     2.443e-01  2.366e-01   1.032   0.3074
## Asian.or.Pacific.Islander.Population   8.317e-02  2.110e-01   0.394   0.6953
## Hispanic.or.Latino.Population          7.952e-04  1.404e-02   0.057   0.9551
## Below.Poverty.Line                    -2.486e-02  3.574e-02  -0.696   0.4903
## Unemployment.Rate                      7.508e-03  5.057e-02   0.148   0.8826
## Police.Officers.per.100.000.Residents -1.105e-03  1.513e-03  -0.730   0.4690
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7788 on 45 degrees of freedom
##   (11 observations deleted due to missingness)
## Multiple R-squared:  0.2322, Adjusted R-squared:  0.06155
## F-statistic: 1.361 on 10 and 45 DF,  p-value: 0.2293
```

Coefficient Tables

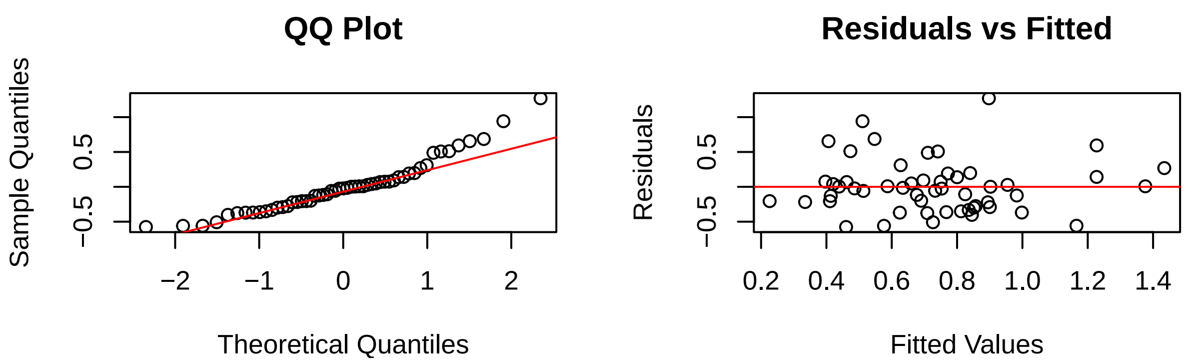
PA Public Defender Residual Diagnostics

Residual Disgnostics Function

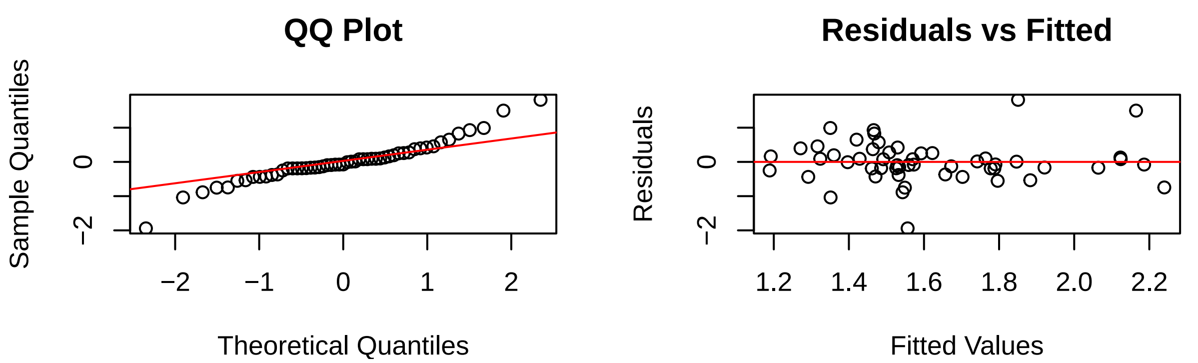
```r
residual_diag_plots(lm_pd_PA)
residual_diag_plots(lm_p_PA)
```

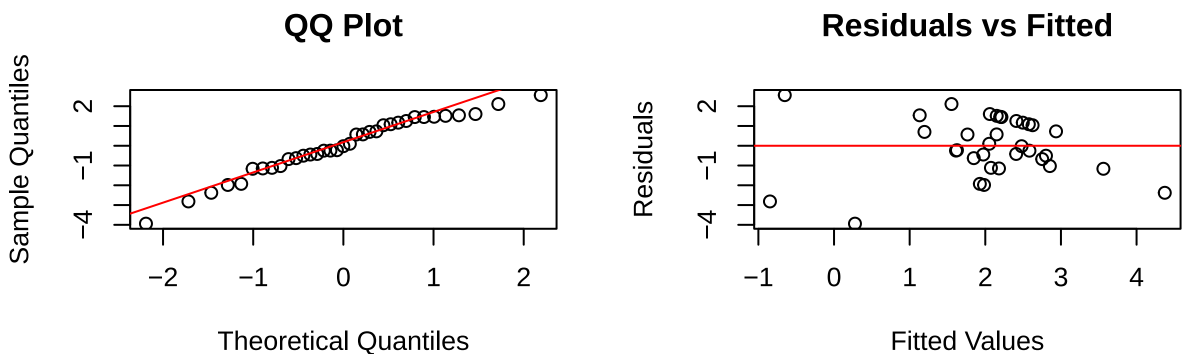



```r
residual_diag_plots(lm_ca_PA)
```

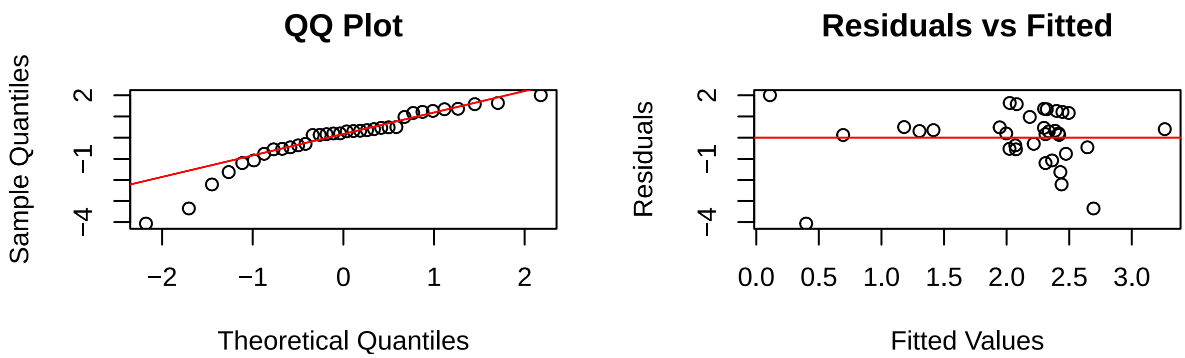



```r
residual_diag_plots(lm_sr_PA)
```

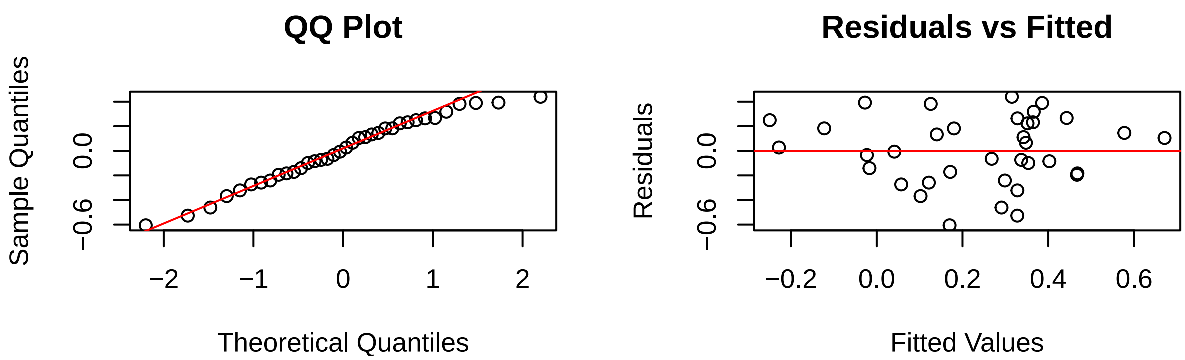




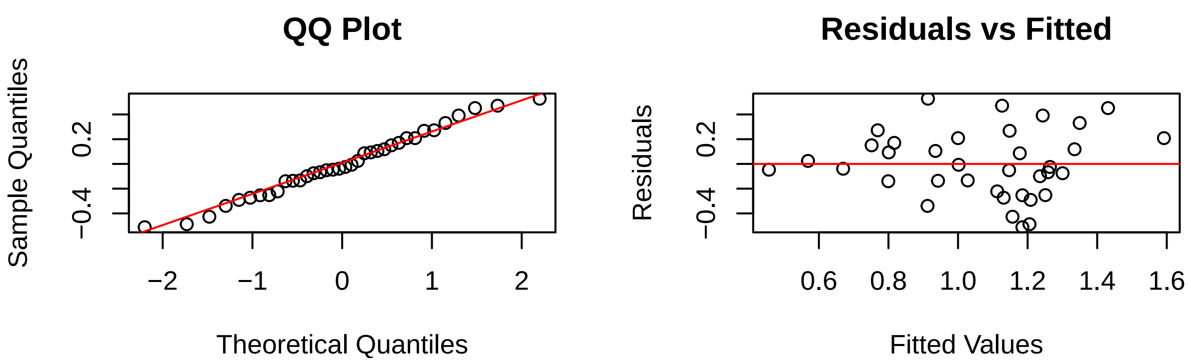

```r
residual_diag_plots(lm_pd_OR)
residual_diag_plots(lm_p_OR)
```

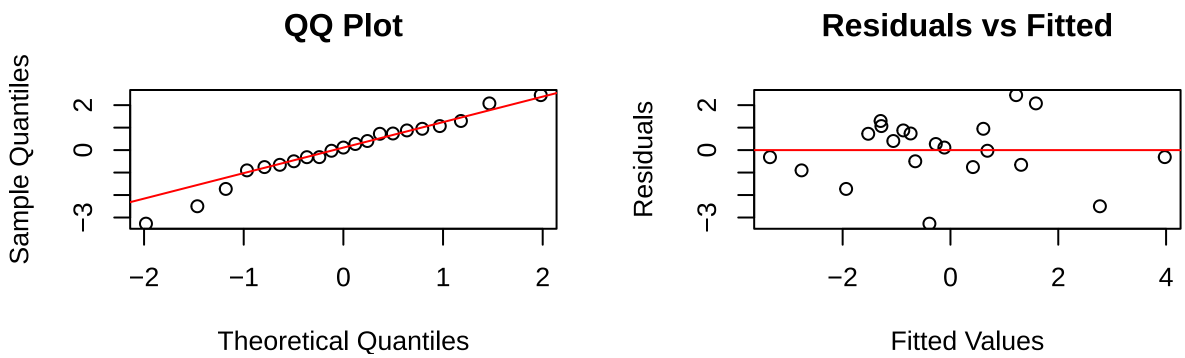



```r
residual_diag_plots(lm_ca_OR)
```

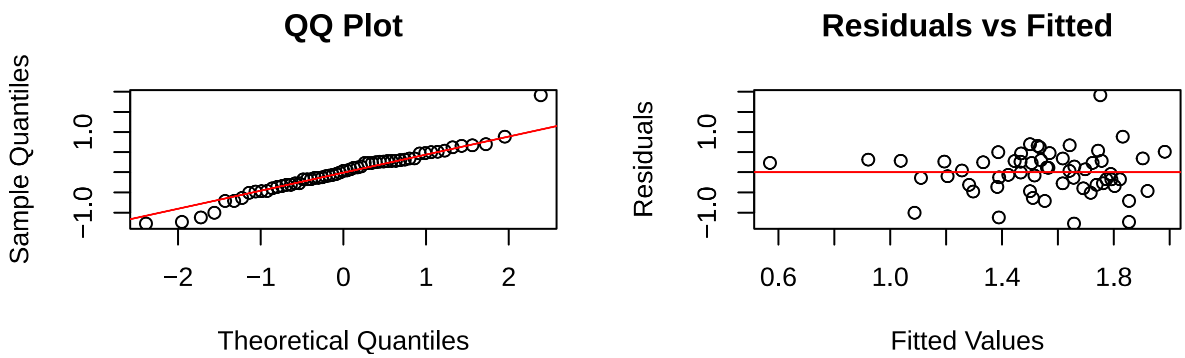



```r
residual_diag_plots(lm_sr_OR)
```

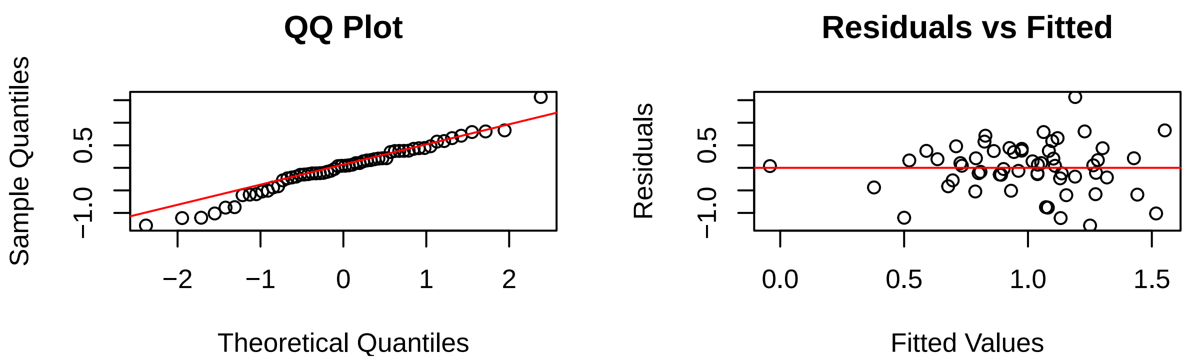




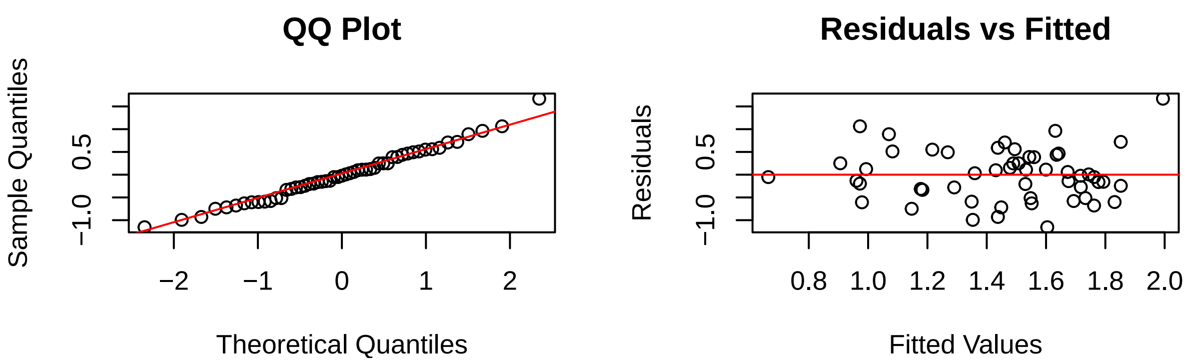

```r
residual_diag_plots(lm_pd_FL)
residual_diag_plots(lm_p_FL)
```

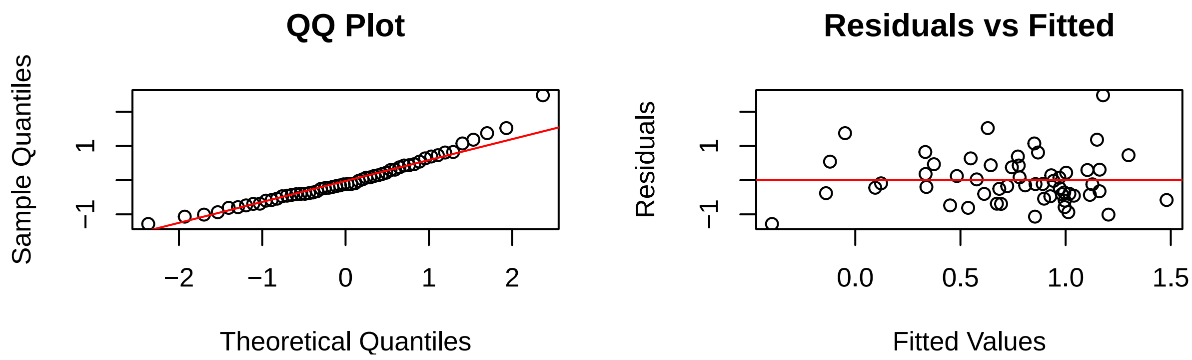



```r
residual_diag_plots(lm_ca_FL)
```




```r
residual_diag_plots(lm_sr_FL)
```

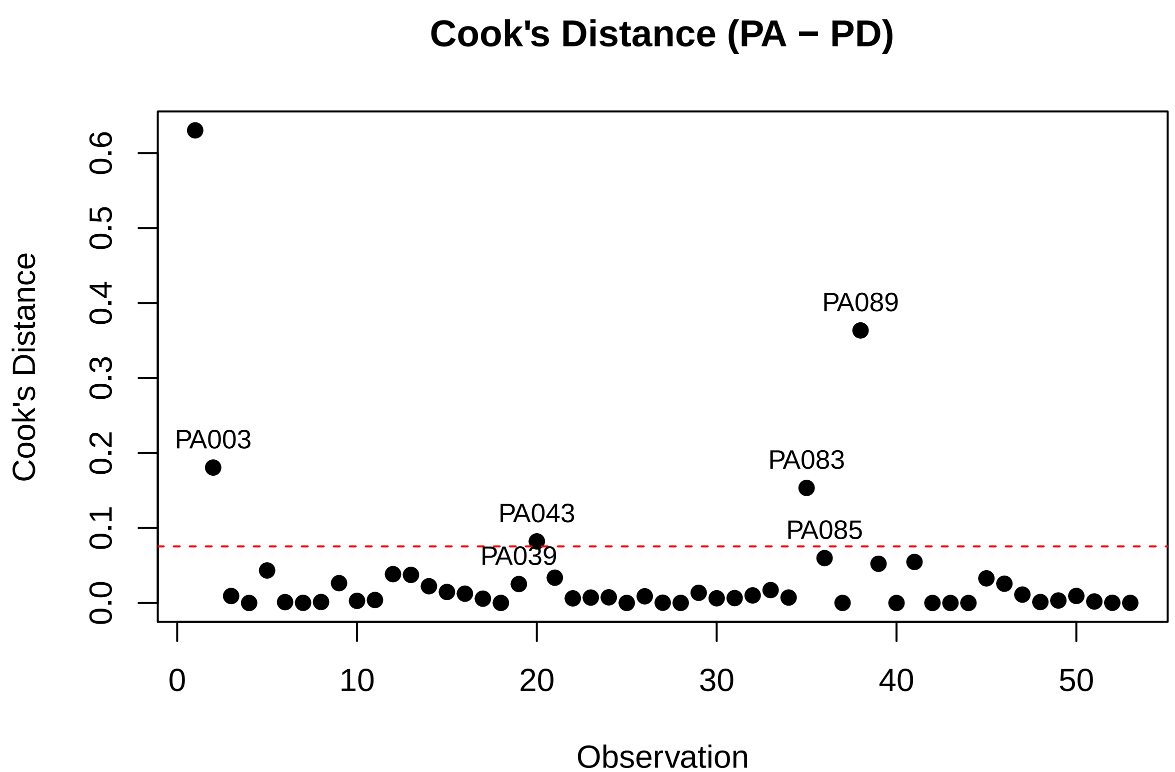




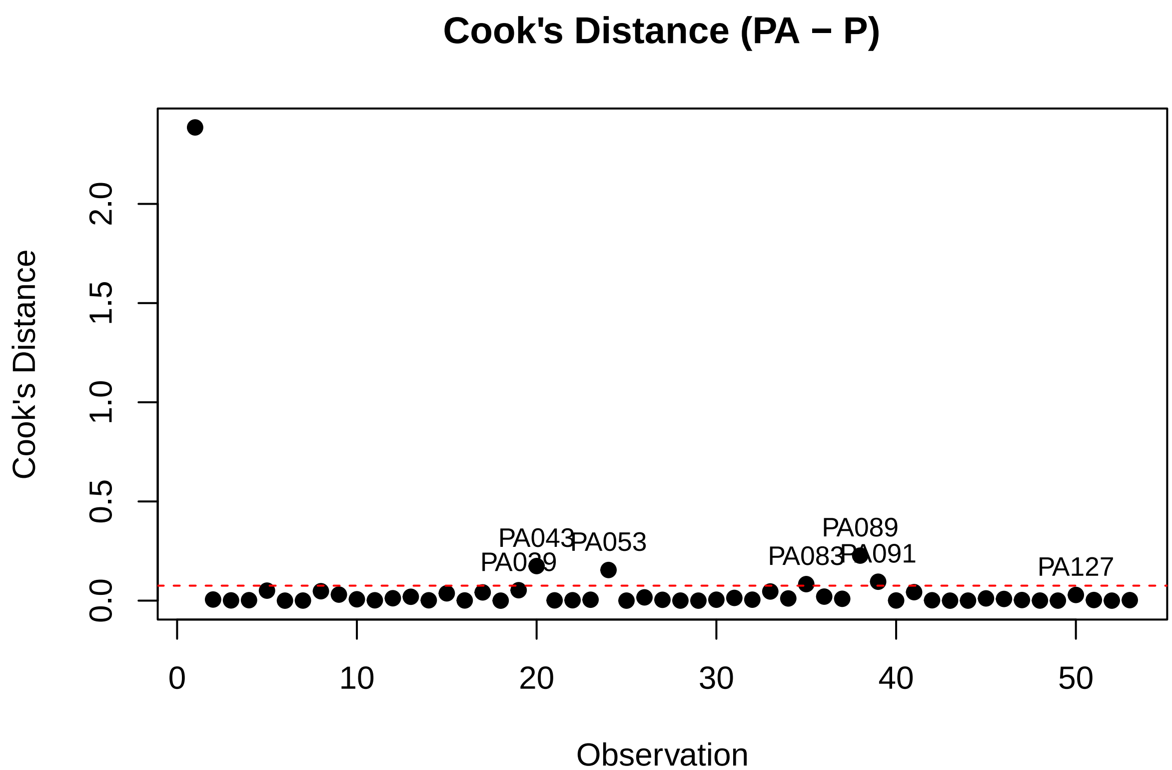
PA Public Defender Leave-One-Out

Cook's Distanse Plot

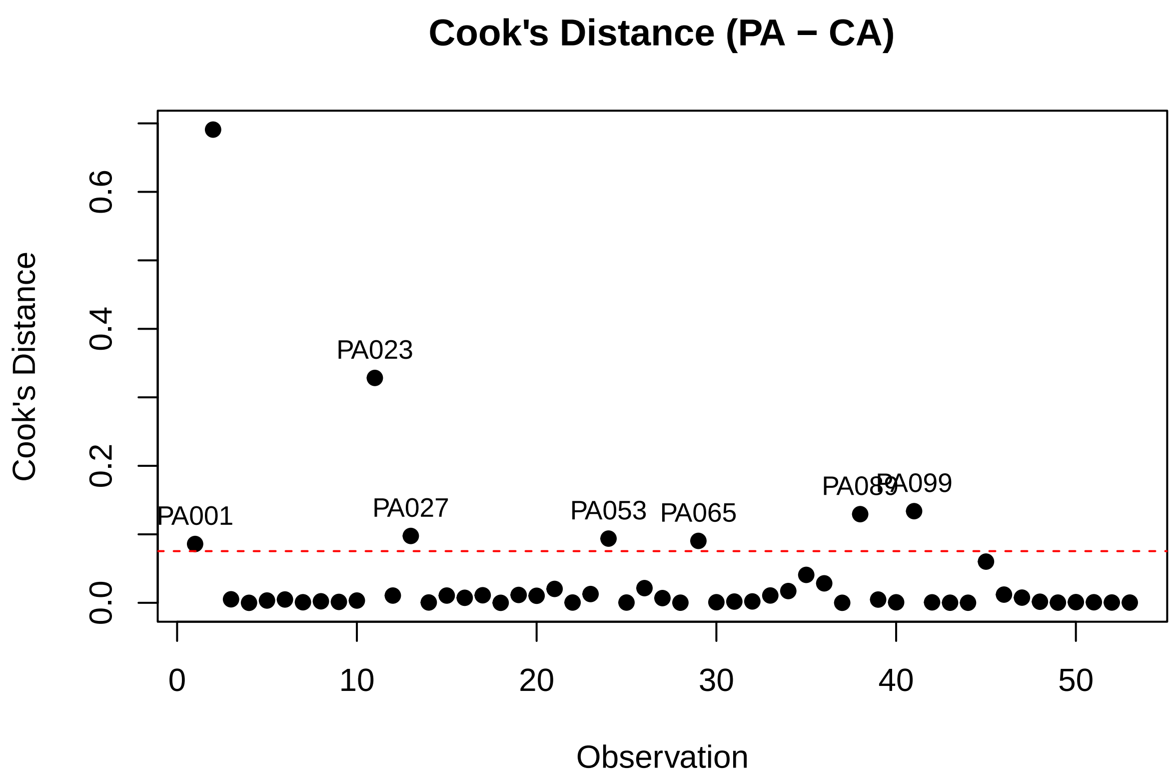
```
cooks_plot(lm_pd_PA, "PA", "PD")
```



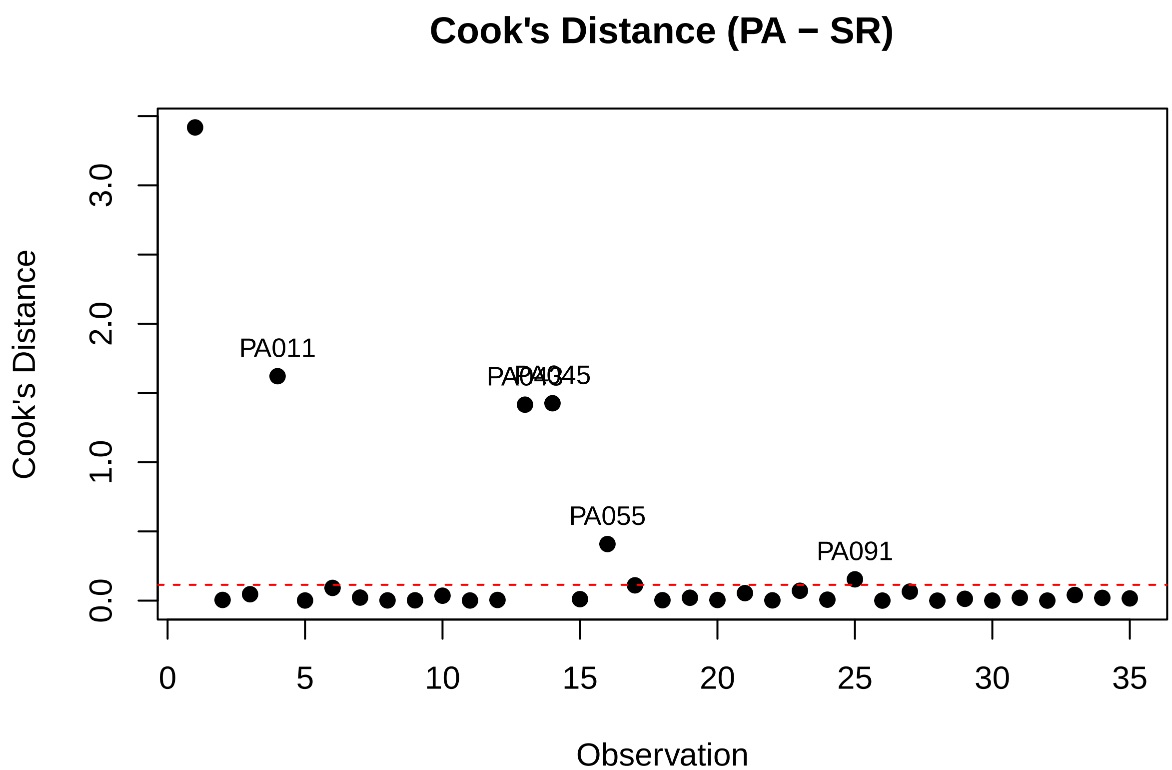

```
cooks_plot(lm_p_PA, "PA", "P")
```



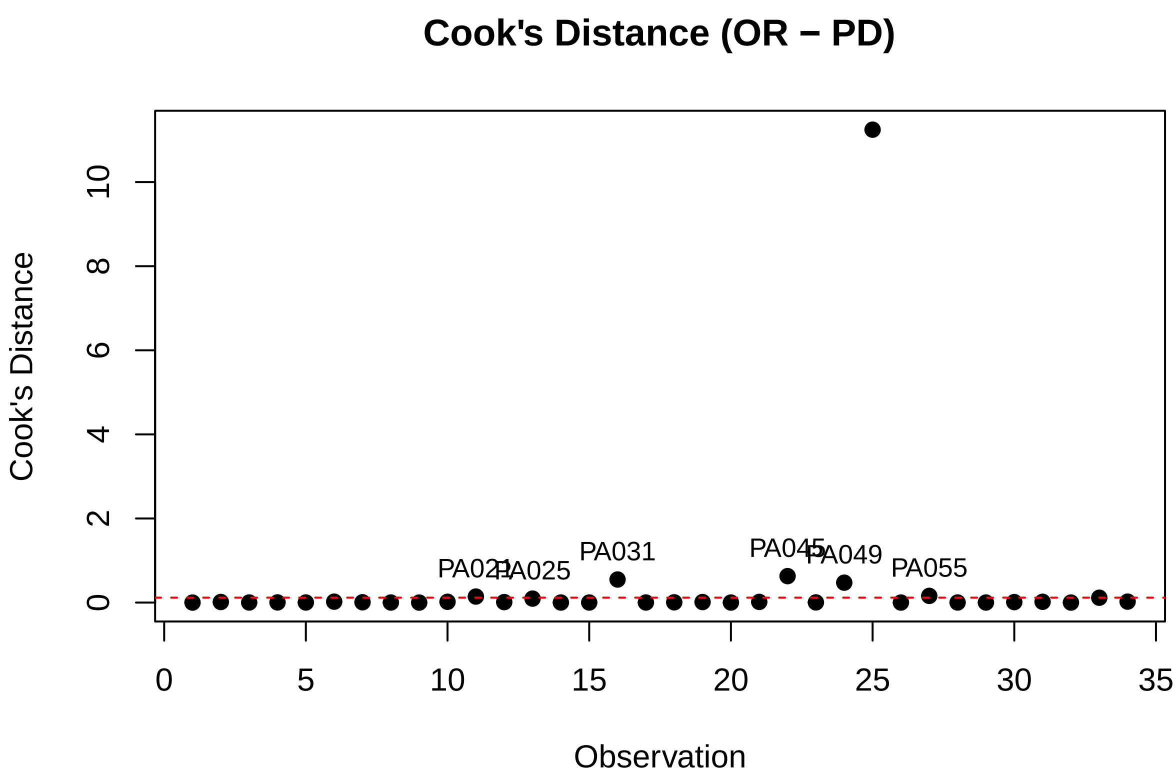

```
cooks_plot(lm_ca_PA, "PA", "CA")
```



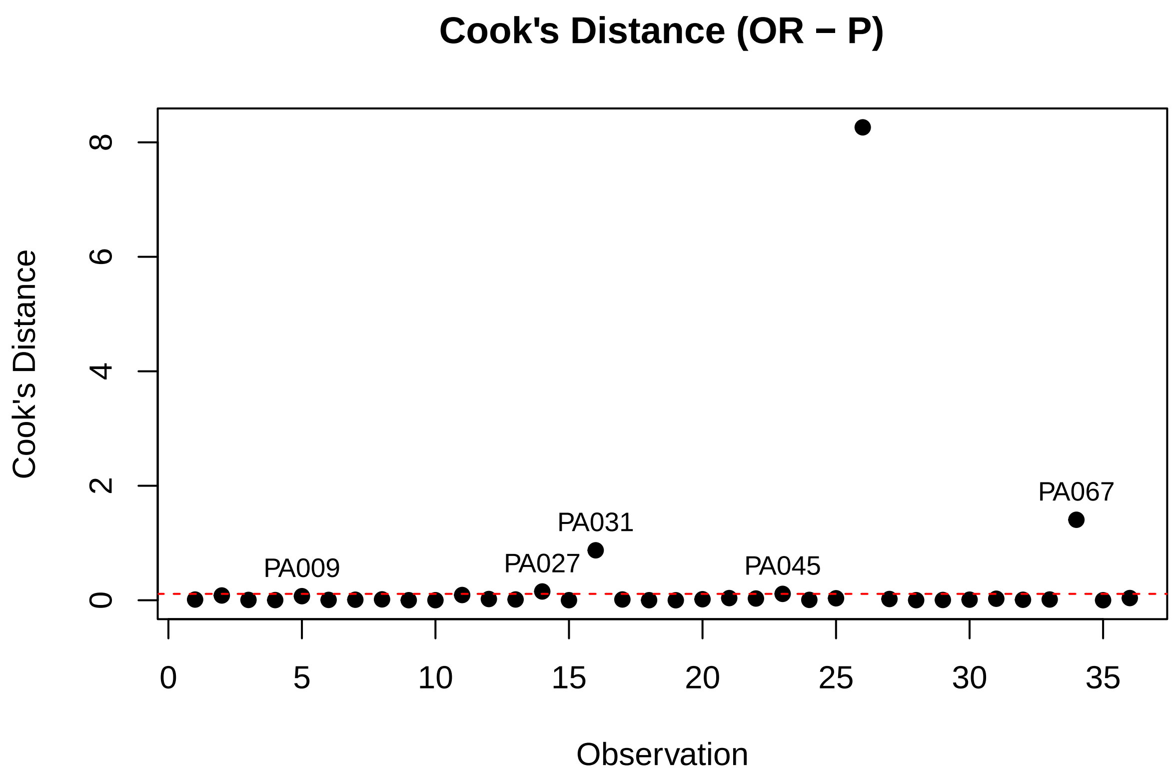

```
cooks_plot(lm_sr_PA, "PA", "SR")
```



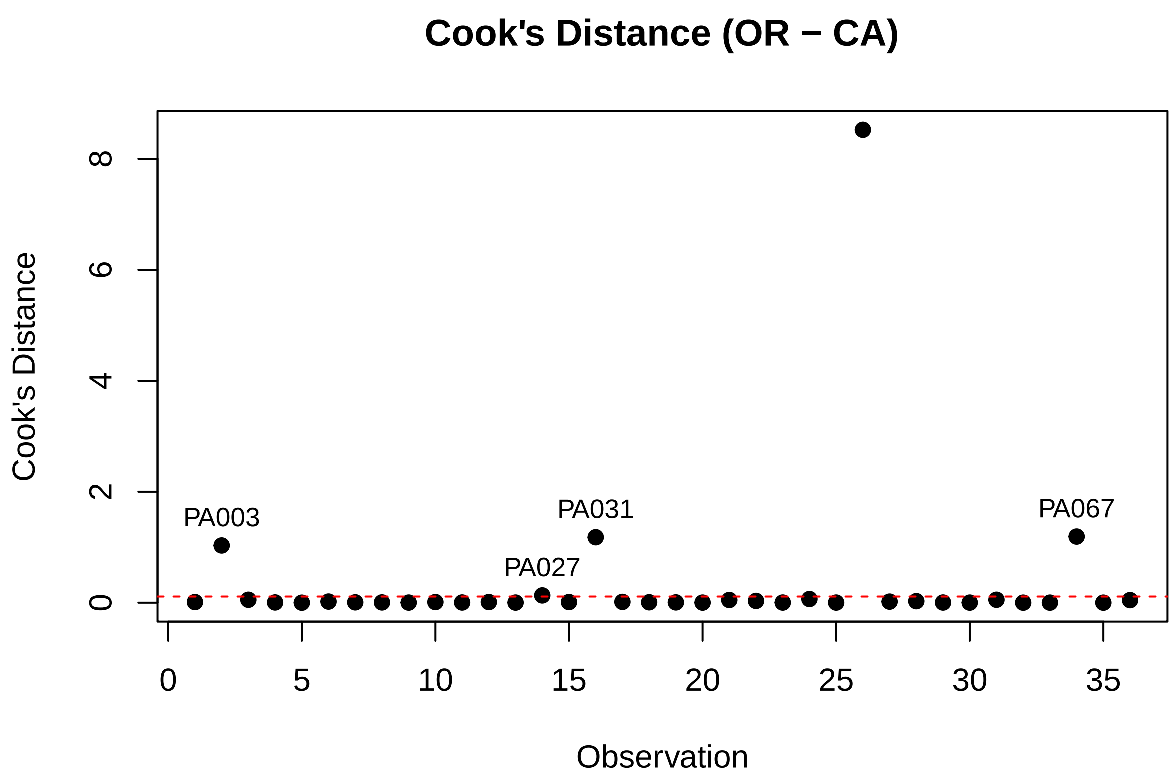

```r
cooks_plot(lm_pd_OR, "OR", "PD")
```



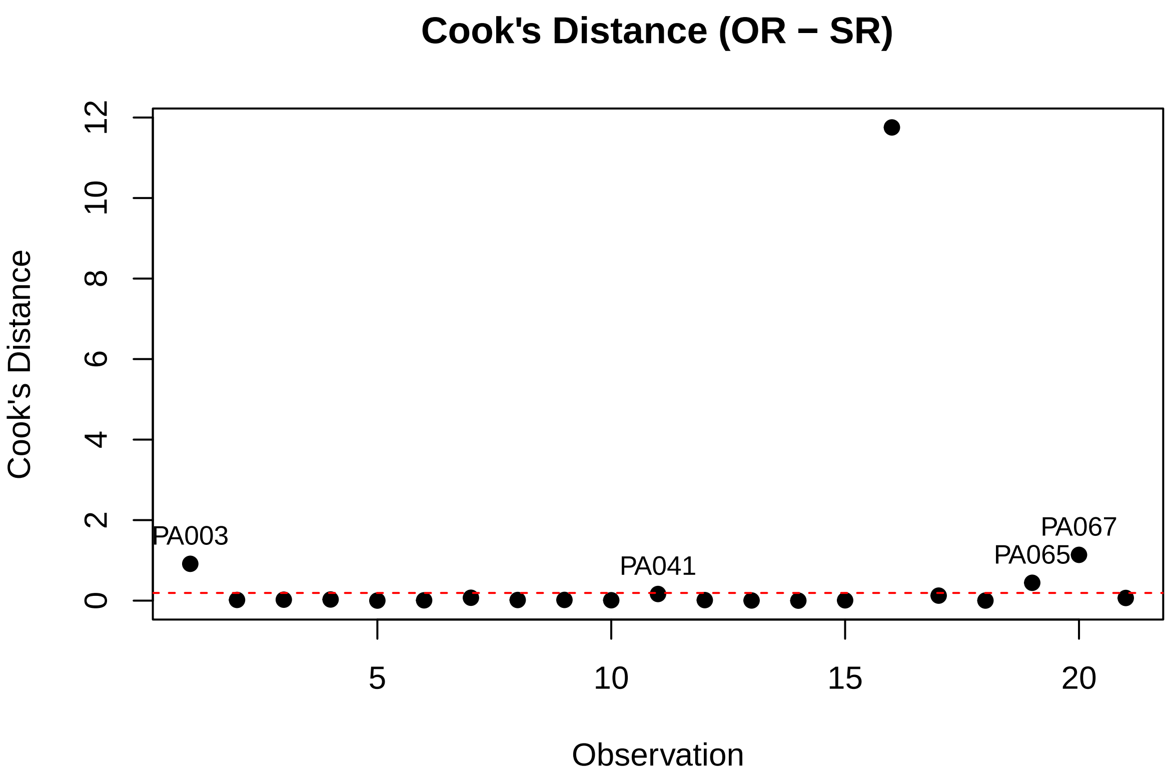

```r
cooks_plot(lm_p_OR, "OR", "P")
```



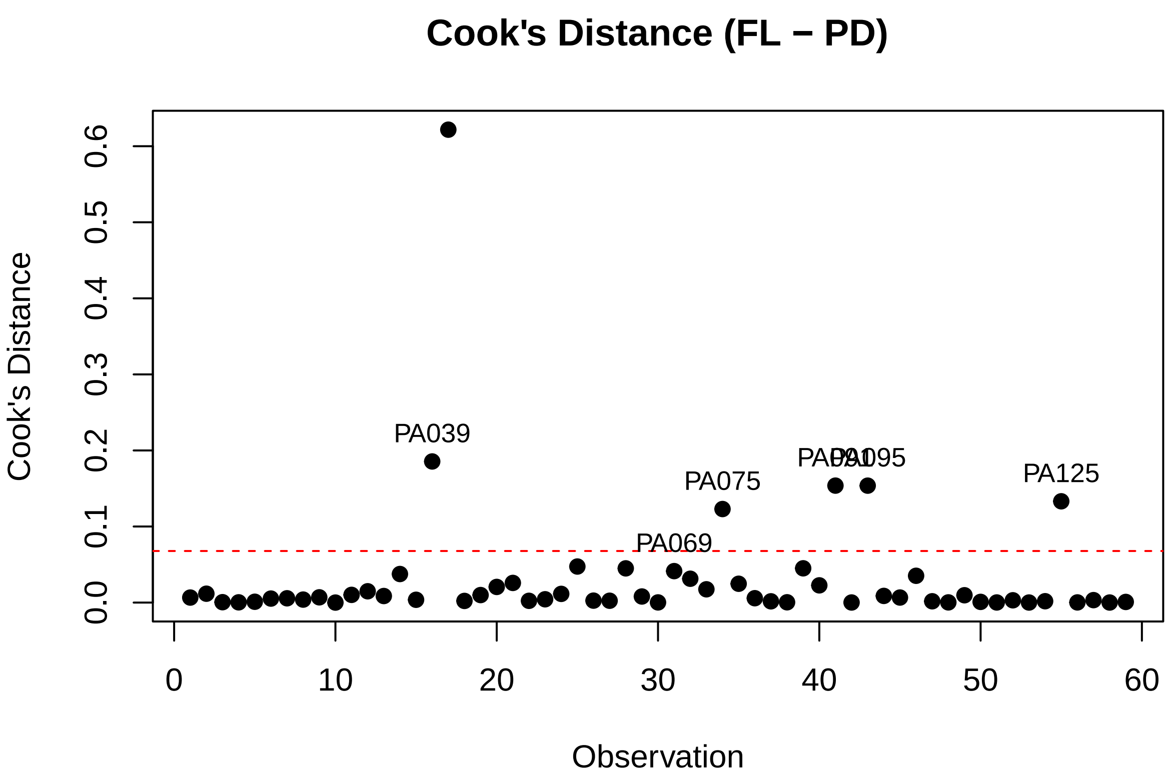

```r
cooks_plot(lm_ca_OR, "OR", "CA")
```



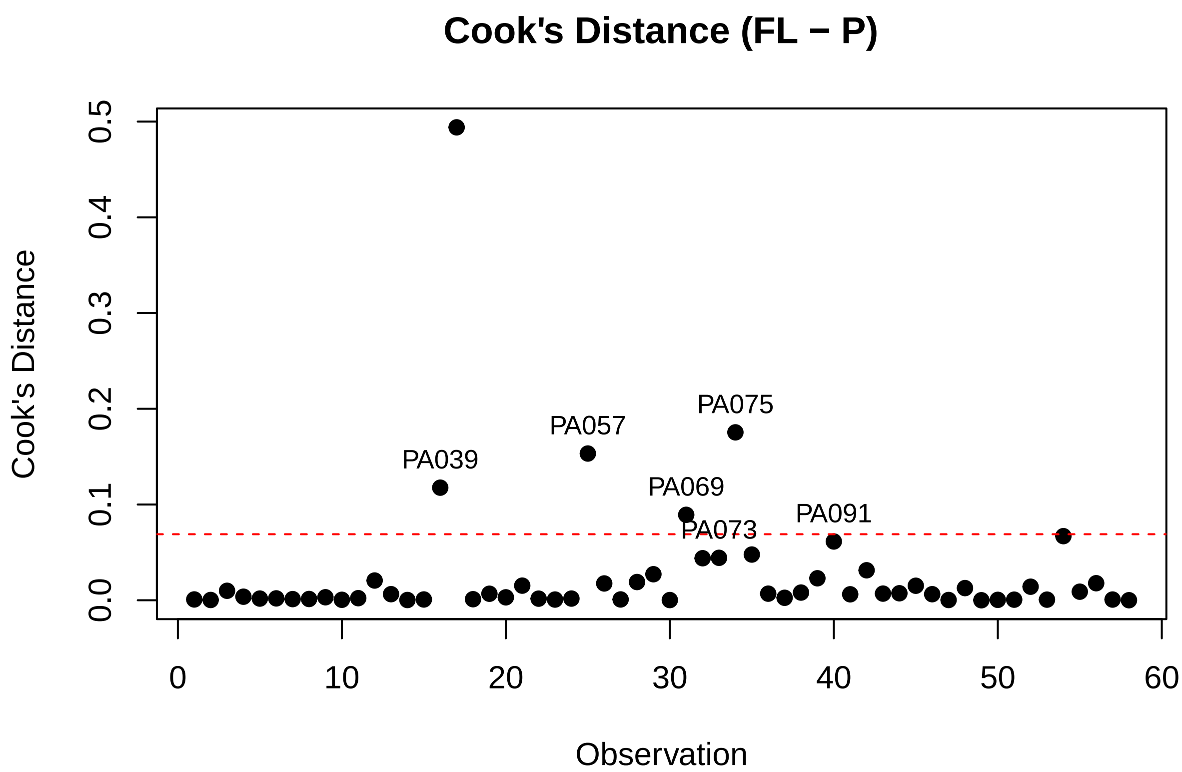

```r
cooks_plot(lm_sr_OR, "OR", "SR")
```



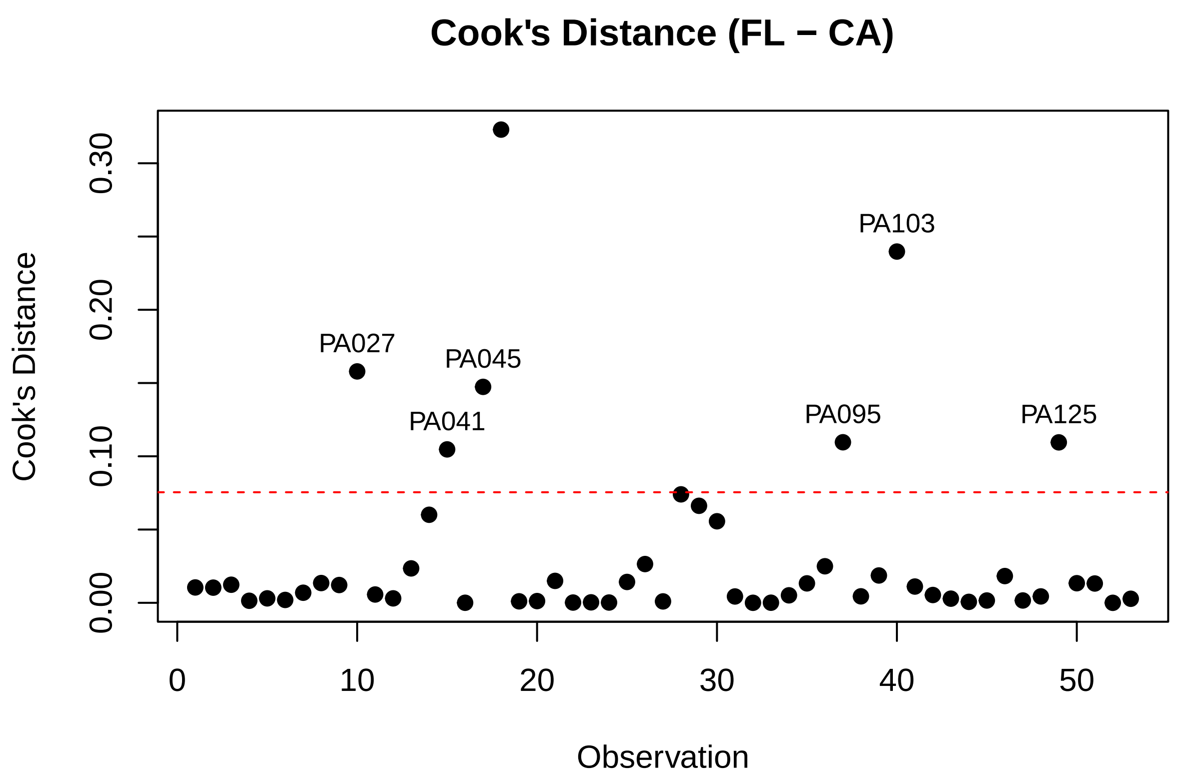

```r
cooks_plot(lm_pd_FL, "FL", "PD")
```



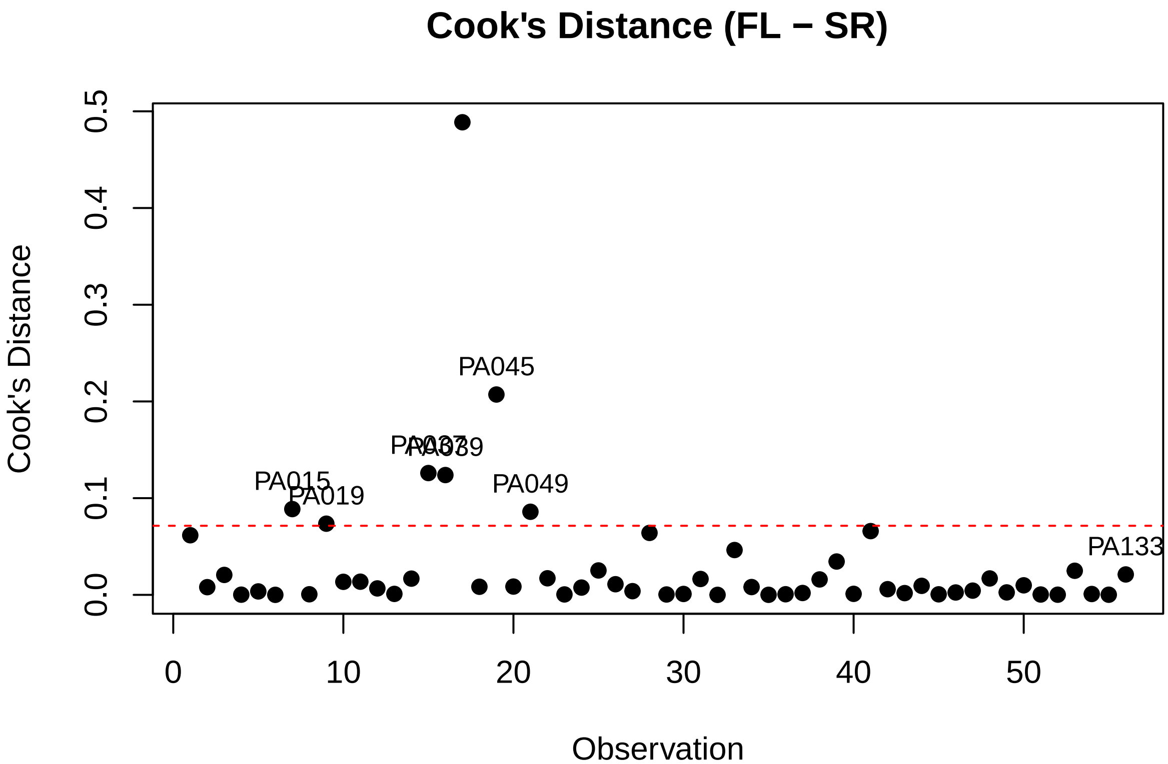

```r
cooks_plot(lm_p_FL, "FL", "P")
```




```r
cooks_plot(lm_ca_FL, "FL", "CA")
```




```r
cooks_plot(lm_sr_FL, "FL", "SR")
```



Ignore below